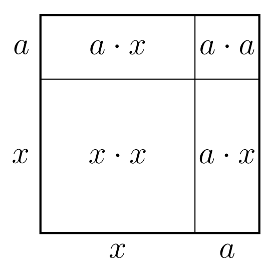
Factoring Special Forms: Squares

Factoring a difference of two squares, factoring a perfect-square trinomial, and taking out a common factor before the special form appears

The two patterns.

- **Difference of squares:** $A^2 - B^2 = (A - B)(A + B)$. Two terms, a minus sign between them, and both terms perfect squares.
- **Perfect-square trinomial:** $A^2 \pm 2AB + B^2 = (A \pm B)^2$. Three terms; the first and last are squares and the middle is twice the product of their roots.
- **Always look for a common factor first.** $2x^2 - 50$ is not a difference of squares as written — it becomes one after the 2 comes out.

The area model used on this sheet. A square of side $x + a$ splits into four regions. No values are shown here — use the numbers given in each problem.



1. Factor completely: $x^2 - 49$

Answer: ________________

2. Factor completely: $x^2 + 10x + 25$

Answer: ________________

3. Factor completely: $9x^2 - 16$

Hint: ask what quantity, when squared, gives $9x^2$.

Answer: ________________

4. Factor completely: $4x^2 - 12x + 9$

Check the middle term against the pattern before you commit to a perfect square.

Answer: ________________

5. A square patio has side length x metres. A square hot tub with side length 4 metres is removed from one corner, so the paved area left over is $x^2 - 16$ square metres.
Write the paved area in factored form.

Answer: ________________

6. Factor completely: $2x^2 - 50$

Two terms and a minus sign, but $2x^2$ is not a perfect square. Deal with that first.

Answer: ________________

7. A square mosaic has an area of $x^2 + 14x + 49$ square centimetres, and its side length is a whole expression in x.

Factor the area to find the side length of the mosaic.

Answer: ________________

8. A student factored $x^2 - 9$ and wrote $(x - 3)^2$.

Expand $(x - 3)^2$ to show what that answer really equals, explain in one sentence which pattern the student confused, then write the correct factorization of $x^2 - 9$.

Answer: ________________

9. Factor completely: $18x^2 - 32$

Neither term is a perfect square as written.

Answer: ____________________

10. Factor completely: $x^4 - 81$

One of the factors is itself a difference of squares. Keep factoring until no factor can be split over the integers.

Answer: ____________________